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A Useful Model for the Calculation of the Performance of Batch and Continuous Jigs

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A model based on the potential energy minimization principle allowing for the dispersive effects due to particle-particle and particle-fluid interactions is developed and applied to the jigging of coal. The model requires only a single parameter, the specific stratification constant. The model generates the equilibrium stratification profile for all components in the jig bed and the separation of each component is then calculated as a function of the total yield of solids. The model is validated against experimental data obtained from a laboratory batch jig and against operating data from several industrial single and two-compartment Baum and Batac jigs. The model calculates the performance of continuous jigs for arbitrary compositions of the feed and is paticularly useful for the reliable simulation of washing plant performance.

Key words: jigging, modeling, simulation, bed stratification, potential energy theory

INTRODUCTION

Theories to describe the mechanism of jigging go back at least as far as 1930 when Gaudin[1] analyzed some of the hydrodynamic aspects of particle behavior in a jig. Lovell *et al.*[2] present a clear short description of the most important theories that have been applied to describe the jigging process. They point out that the main shortcomings of the hydrodynamic theory of Gaudin[1] and potential energy theory of Mayer[3] is the neglect of the strong particle-particle interactions that dominate the behavior of the jig bed. Lyman[4] has recently provided a comprehensive review of the jigging process and he showed that several particular features can be described but the theoretical models have not yet evolved to a state that allows them to be used for the prediction of the actual performance of industrial jigging operations.

In this paper we attempt to rectify this state of affairs and take the view that the separating characteristics of a jig are dominated by the stratification that occurs in the jig bed. A realistic model for stratification is proposed and a useful model that describes industrial jigs is consequently developed. The resulting model is unique in that it is not based on empirically determined partition curves to describe the separation that is achieved in the jig and the model has the added advantage of requiring only a single parameter to describe the stratification process. In the absence of any useful theory that describes the jigging process, various researchers have, in the past, relied on fairly comprehensive sets of experimental data to develop empirical models. The most successful of these is the partition curve approach that was described as early as 1937 by Tromp[5] and partition curves have become the standard method for the calculation of coal washing unit performance[6]. The partition curve approach can be very effective for gravity separation units that use a dense medium to achieve separation of the particles but the method is less satisfactory for autogenous gravity separation operations which rely on bed stratification to achieve separation. The term autogenous gravity separation was introduced by Vetter[7] to distinguish those processes in which the separation occurs in a density environment that is created by the separating particles themselves from those that operate within a liquid medium environment that has a definite imposed density. The important autogenous gravity separation operations are the water-only cyclone, the pinched sluice, the Reichert cone and the jig. It has often been asserted in the literature that the autogenous separators are equivalent to medium-based separators with water being regarded as the medium. This is erroneous. In autogenous gravity separators, the separation takes place in a density environment created by the particles themselves which create their own medium and the density of the working fluid does not effectively control the cutpoint.

The main shortcoming of the partition curve method when applied to autogenous separators is the unpredictable variation of the partition curve with variation in the washability of the feed material. The partition curve for a jig is also very dependent on the physical position of the cutter or discharge gate. In fact, it is impossible to accurately predict the partition curve for a jig from "typical" curves for the unit since the curve depends so strongly on the characteristics of the feed material and the physical setup of the jigging equipment. In spite of these shortcomings, the partition curve method, usually applied in some generalized form[8] is still the basis on which most coal preparation engineers rely to predict the steady-state operation of coal jigs and the practice is very common in the coal preparation industry.

A more direct approach is taken in this paper. The partition curve, and therefore the separation performance of the jig, is calculated as a result of the stratification of the bed that occurs because of the jigging action. Thus the stratification density profile is calculated using the actual composition of the feed material and the separation characteristics are determined by cutting off the top layer of the bed as the clean coal product.

The method used to calculate the stratification profile is based on the potential energy theory of Mayer[3] who showed that under ideal conditions a bed of particles will stratify in such a way as to minimize the gravitational potential energy of the bed. In practice this ideal stratification is never achieved because various dispersive forces are always present which tend to disperse the particles by random motions and thus destroy the ideal stratification profile. The actual stratification achieved represents a balance between the stratification forces and the dispersive forces. The theory based on these ideas to calculate the actual equilibrium stratification profile was developed by King[9] and has already been completely worked out for binary systems. However, the binary system is not useful for real coal washing operations which require a model that can describe the behavior of many different particles of variable density. This paper extends the earlier binary theory to the practically important multicomponent systems. An efficient solution procedure to the equilibrium stratification equations for the multicomponent case is described and a framework is developed by which the industrial jigging of coal can be effectively modeled.

STRATIFICATION OF MULTICOMPONENT MIXTURES IN A BATCH JIG

Consider a packed bed of monosize spherical particles whose densities vary over a wide range. The gradient of the potential energy of a particle of density ρ in a bed of particles of average density $\bar{\rho}$ is given by[9]

$$\frac{dE}{dH} = gV_p(\rho - \bar{\rho}) \tag{1}$$

where V_p is the particle volume, E is the gravitational potential energy, H is the height of the particle in the bed measured from the bedplate. The average density of the particles, $\bar{\rho}$, at height H is given by

$$\bar{\rho} = \int_0^\infty \rho\, C_\rho\, d\rho$$

where C_ρ is the volumetric concentration of particles within the solid phase with density ρ at height H. The potential energy gradient causes a particle to migrate upward or downward in the jigged bed depending on the sign of $\rho - \bar{\rho}$. If $\rho > \bar{\rho}$ the particle will move down and *vice versa*. The rate at which particles move relative to the bed is proportional to the potential energy gradient so that the flux of particles of density ρ due to the potential energy gradient is given by

$$n_S = -C_p u \frac{dE}{dH} \tag{2}$$

where n_S is the stratification flux and u is the specific mobility of the particle which is defined as the penetration velocity achieved by a particle in the absence of any dispersive forces under a unit potential energy gradient. The specific mobility would, for example, be 1000 times the average velocity at which a 1-cm spherical particle of density 4610 kg/m^3 would sink in a jigged bed of similar particles all having density 2700 kg/m^3. The significance of the relative velocity between a particle and the stratifying bed was already recognized by Gaudin[1] but has been neglected by most researchers since the 1930's. Substituting (1) in (2)

$$n_s = C_\rho\, ugV_p(\rho - \bar{\rho}) \tag{3}$$

Opposing the stratification flux, there is the diffusive flux due to particle-particle and particle-fluid interactions. It is described by a Fickian equation of the type

$$n_D = -D \frac{dC_\rho}{dH} \tag{4}$$

The diffusion constant D in Equation (4) is dependent on the particle size, shape and bed expansion mechanism.

The ideal Mayer stratification profile that minimizes the potential energy of the bed will never be achieved in practice. A dynamic state of equilibrium of the bed will be established when the stratification flux for each particle type is exactly balanced by the corresponding diffusive flux. This condition is defined by

$$n_D = -n_S \tag{5}$$

so that

$$\frac{dC_\rho}{dH} = -\frac{ugV_p}{D} C_\rho (\rho - \bar{\rho}) \tag{6}$$

Defining the relative height by $h = \dfrac{H}{H_b}$, where H_b is the bed height and the specific stratification constant as $\alpha = \dfrac{ugV_pH_b}{D}$ Equation (6) becomes

$$\frac{dC_\rho}{dh} = -\alpha C_\rho [\rho - \bar{\rho}(h)], \tag{7}$$

α quantifies the stratification action of the jig and it should be independent of particle density. In a monosize bed of particles having similar shapes all particles have the same value of α. The solution to differential Equation (7) gives the vertical concentration profile of monosize particles of density ρ. No boundary conditions can be specified *a priori* for Equation (7) because it is not possible to specify the concentration of any species at either the top or bottom of the stratified bed. Solutions to Equation (7) must satisfy the conditions

$$\int_0^1 C_\rho \, dh = C_\rho^f \qquad \text{for all } \rho \tag{8}$$

$$\int_0^\infty C_\rho \, d\rho = 1 \qquad 0 \le h \le 1 \tag{9}$$

where C_ρ^f is the feed density distribution by volume. Equations (8) and (9) are sufficient to define the solution to Equation (7) without any other boundary conditions.

In practice, the feed density distribution is determined by float-sink tests, which classify the feed into a finite number of density classes. Thus a more convenient, density discretized form of the model for n components is expressed by

$$\frac{dC_i(h)}{dh} = -\alpha C_i(h) [\rho_i - \bar{\rho}(h)] \qquad i = 1,2,\ldots n \tag{10}$$

where $C_i(h)$ is the volume fraction at height h of material characterized by density ρ_i. The representative density of each interval, ρ_i, corresponds to the mean density of each class. Considering the lack of information at both ends of the density spectrum, particularly of the last sinks fraction, the mean densities of those intervals are usually only known approximately.

The average solids density at level h is given by

$$\bar{\rho}(h) = \sum_{i=1}^{n} C_i(h)\,\rho_i \tag{11}$$

The system of equations defined by Equations (10) must be solved subject to

$$\sum_{i=1}^{n} C_i(h) = 1 \qquad 0 \le h \le 1 \tag{12}$$

$$C_i^f = \int_0^1 C_i(h)\,dh \tag{13}$$

This system of differential equations reflects the autogenous nature of the stratification process since the local gradient of the stratification profile depends on the local bed properties and the entire bed profile is compatible with the make-up of the feed through Equation (13).

In order to determine the equilibrium concentration profile it is necessary to solve the system of n coupled differential equations defined by Equations (10) subjected to Equations (12) and (13).

An analytical solution for the particular case $n = 2$ (binary mixture) was developed by King[9]. For multicomponent mixtures, analytical solutions are not available. In that case the equilibrium stratification equations must be integrated numerically.

One obvious approach to the solution of the system of Equations (10) is to assume a set of $C_i(0)$, integrate Equations (10) numerically and check the resulting profiles against Equation (13). Using an optimization procedure, the initial guesses of $C_i(0)$ are refined and the procedure is repeated. This procedure, however, converges only slowly and exhibits weak stability characteristics.

A very efficient procedure for integrating the equilibrium equations was developed using the following iterative method.

Integration of the stratification Equations (10) from the bottom of the bed with the unknown initial condition $C_i(0) = C_i^0$ gives

$$C_i(h) = C_i^0 \exp[-\alpha\rho_i h + \alpha \int_0^h \bar{\rho}(u)du] \tag{14}$$

Substituting Equation (14) into Equation (13)

$$\int_0^1 C_i^0 \exp[-\alpha\rho_i h + \alpha \int_0^h \bar{\rho}(u)du]dh = C_i^f \tag{15}$$

and rearranging gives an estimate of the concentration of each species at the bottom of the bed

$$C_i^0 = \frac{C_i^f}{\int_0^1 \exp[-\alpha\rho_i h + \alpha \int_0^h \bar{\rho}(u)du]dh} \tag{16}$$

An initial average density profile $\bar{\rho}(h)$ is assumed, each C_i^0 is calculated using Equation (16) and normalized to satisfy

$$\sum_{i=1}^{n} C_i^0 = 1 \tag{17}$$

New concentration and average density profiles are calculated from Equations (14) and (11) and the procedure is repeated until convergence.

This iterative procedure is very efficient and robust, capable of rapidly generating solutions for any number of components encountered in practice and the algorithm converges to the solution of Equations (10).

The solution to Equations (10) for a hypothetical four-component mixture is illustrated in Figure 1, which shows a typical equilibrium stratification profile and compares it to the ideal Mayer profile that would be achieved without dispersion.

The mass yield of solids in the clean coal product obtained by slicing the bed at a relative height h_s is given by

$$Y(h_s) = \frac{\int_{h_s}^1 \bar{\rho}(h)dh}{\int_0^1 \bar{\rho}(h)dh} \tag{18}$$

The recovery of the ith density component to the clean coal product at a height h_s is given by

$$R_i(h_s) = \frac{\int_{h_s}^1 C_i(h)dh}{\int_0^1 C_i(h)dh} \tag{19}$$

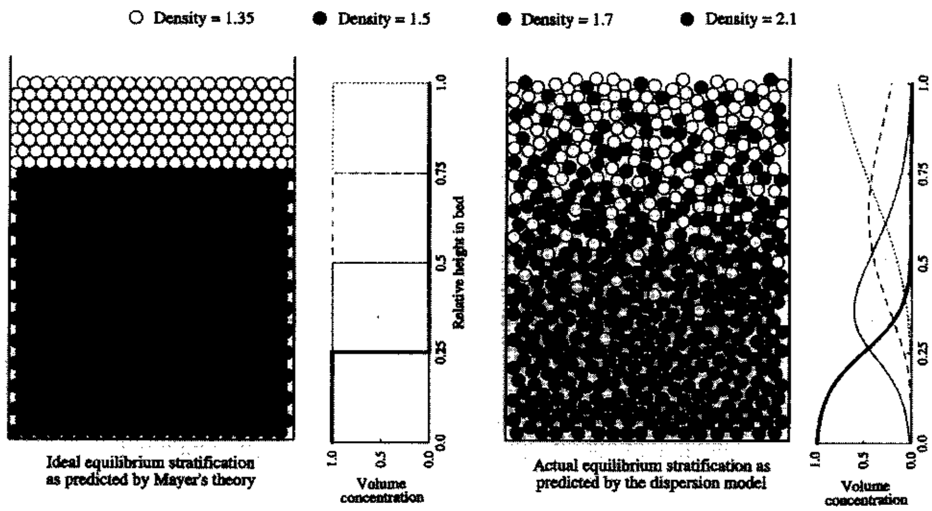

FIGURE 1 The effect of particle dispersion on the equilibrium stratification profile in a batch jig. The beds represent accurate simulations using the ideal Mayer theory and the dispersion model. The graphs next to each bed show the calculated vertical concentration profiles for the four components. A value of $\alpha = 0.03$ was used to calculate the concentration profiles generated by the dispersion model.

Specific stratification constant values range from zero for a perfectly mixed bed to infinity for perfect stratification. If specific density units are used in kg/m^3, typical values of alpha range from 0.001 for a very poor separation to 0.5 for an exceedingly accurate separation. Although independent of the washability distribution of the feed, the specific stratification constant is dependent on the size and shape of the particles and on operating conditions used in the jig.

STRATIFICATION OF MULTICOMPONENT MIXTURES IN A CONTINUOUS JIG

Three distinct subprocesses can be identified in a continuous jig: bed stratification produced by the vertical pulsation of the bed, longitudinal transport of the bed due to water flow along the axis of the jig and splitting of the bed into product and refuse layers.

It is known that a vertical velocity profile exists in the continuous jig[10–12]. In each compartment of the jig the various bed layers move at different velocities, with the upper layers generally moving faster than the lower ones. The velocities of particles at the end of the bed depend primarily on the velocities of the layers to which they report. The velocity profile at the discharge end of the bed determines the discharge rate of each layer as it leaves the jig. Equation (8) for the batch jig must be replaced by Equation (20) for the continuous jig.

$$C_\rho^f = \frac{\int_0^1 C_\rho(h)V(h)dh}{\int_0^\infty \int_0^1 C_\rho(h)V(h)dh\, d\rho} \tag{20}$$

where $V(h)$ is the longitudinal velocity of the bed at height h.

The velocity profile must be known in order to be able to integrate Equation (20). The actual longitudinal velocity at each bed height can only be determined experimentally and depends on the operating conditions in the jig. Defining a dimensionless velocity by

$v(h) = \dfrac{V(h)}{V_{max}}$ and discretizing Equation (20) we get

$$C_i^f = \frac{\int_0^1 C_i(h)v(h)dh}{\sum_{j=1}^{n} \int_0^1 C_j(h)v(h)dh} \tag{21}$$

The continuous jigging model for monosize feeds consists of Equations (10) subjected to the integral conditions given in Equations (21) and (11). Let the denominator of Equation (21) be equal to a constant, say $1/\beta$, then

$$C_i^f = \beta \int_0^1 C_i(h)v(h)dh \tag{22}$$

A similar iterative solution procedure to the one used in the batch model is used for the continuous model. Then substituting Equation (14) into Equation (22) and rearranging,

$$\beta C_i^o = \frac{C_i^f}{\int_0^1 \exp[-\alpha\rho_i h + \alpha \int_0^h \bar{\rho}(u)\, du]v(h)dh} \tag{23}$$

βC_i^o is determined iteratively and C_i^o is obtained from

$$C_i^o = \frac{\beta C_i^o}{\sum_{j=1}^{n} \beta C_j^o} \tag{24}$$

In the jigging process, particularly for well operated jigs, the longitudinal velocity is expected to increase monotonically from the bottom to the top of the bed, as observed by Rong *et al.*[11] and Lyman *et al.*[12] The limited amount of direct measurements of velocity profiles found in the literature, however, does not allow the development of precise models for the velocity profile. A simple but effective empirical model for the velocity profile which does not appear to conflict with the available experimental data is used in this paper and is given by

$$v(h) = \kappa h + (1-\kappa)h^2 \tag{25}$$

where κ is a parameter of the model.

In a continuous jig, the separation of stratified layers is affected by refuse and middlings discharge mechanisms. Additional misplacement of clean coal and refuse is believed to happen as a result of the turbulence produced by the operation of these removal systems[2]. The absence of any quantitative experimental work on this effect and the wide variety of refuse ejectors encountered in practice does not allow the precise modeling of this subprocess at this stage. A simple model which considers that the turbulence produced by splitting the stratified layers creates a perfectly mixed region of relative thickness 2δ is proposed tentatively here. The yield of solids to the clean coal product can now be calculated by

$$Y(h_s,\delta) = \frac{\dfrac{1}{2}\displaystyle\int_{h_s-\delta}^{h_s+\delta} \bar{\rho}(h)v(h)dh + \int_{h_s+\delta}^{1} \bar{\rho}(h)v(h)dh}{\displaystyle\int_{0}^{1} \bar{\rho}(h)v(h)dh} \tag{26}$$

The recovery to the clean coal product of the ith density component is given by

$$R_i(h_s,\delta) = \frac{\dfrac{1}{2}\displaystyle\int_{h_s-\delta}^{h_s+\delta} C_i(h)v(h)dh + \int_{h_s+\delta}^{1} C_i(h)v(h)dh}{\displaystyle\int_{0}^{1} C_i(h)v(h)dh} \tag{27}$$

The density distribution of the clean coal product by volume is given by

$$C_i^p(h_s,\delta) = \frac{C_i^f R_i(h_s,\delta)}{\displaystyle\sum_{j=1}^{n} C_j^f R_j(h_s,\delta)} \tag{28}$$

Using the same four-component mixture as in the batch model (Figure 1), the continuous jigging model showing the velocity profile and the splitting mechanism is illustrated in Figure 2. A thick layer of refuse accumulates in the bottom of the bed as a result of the longer residence times experienced by the heavier material in the jig. The concentration of the separate components in the product streams is governed by Equation (28) and differs significantly from the equilibrium concentration in the bed itself.

This model can also be used for simulating separations in multiple compartment jigs. The first compartment is modeled as a single compartment jig and the second compartment is considered to be a single compartment with the feed equal to the light product from the first compartment.

This is illustrated schematically in Figure 3. Thus Equation (21) becomes

$$C_i^P = \frac{\int_0^1 C_i^c(h) v_2(h) dh}{\sum_{j=1}^{n} \int_0^1 C_j^c(h) v_2(h) dh} \tag{29}$$

where C_i^P is the density distribution in the light product from the first compartment and $C_i^c(h)$ and $v_2(h)$ are the equilibrium concentration and velocity profiles in the second compartment, respectively.

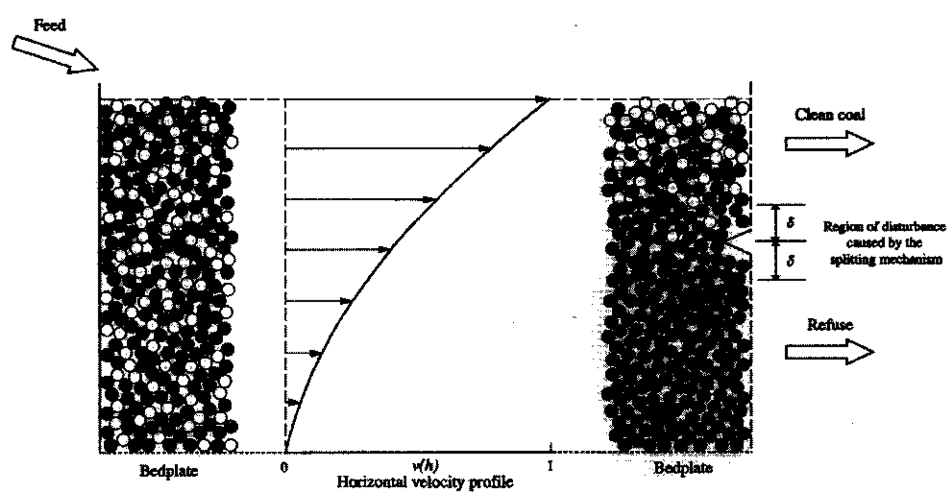

FIGURE 2 Schematic representation of the continuous jig showing the horizontal velocity profile and the splitting mechanism. The feed has four components in equal amounts as in Figure 1 and a value of κ = 0.3 was used to integrate the model. Note the build up of the heaviest component in the jig compared with the stratification profile for the batch jig in Figure 1.

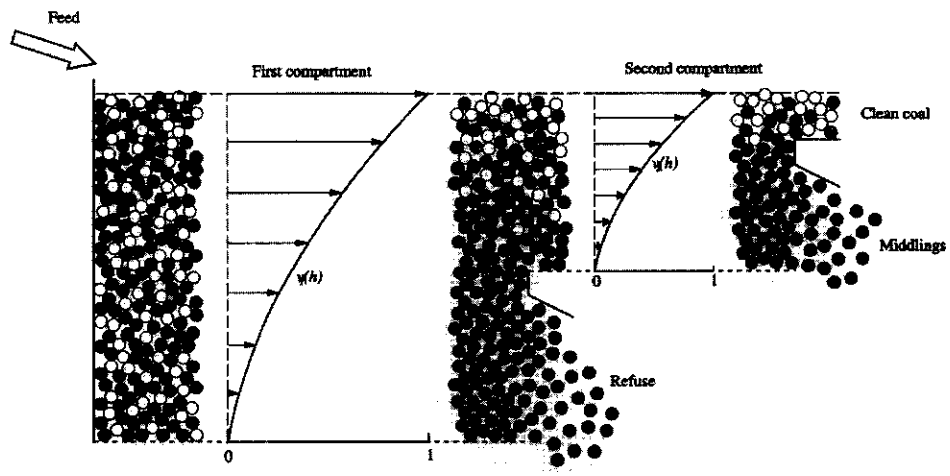

FIGURE 3 Schematic representation of the two-compartment jig.

The dispersion model (10) for the second compartment is integrated subjected to Equation (29) in a manner similar to that described earlier to generate the equilibrium stratification profile.

APPLICATION OF THE MODEL TO OPERATING JIGS

Comparison of the Model to Batch Jigging Data

Application of the model to binary mixtures of monosize particles has been presented earlier by King[9] and compared to experimental data. The multicomponent stratification model is tested here using data collected by Vetter[7] for batch jigging of ternary mixtures of PVC cubes and of closely sized coals and crushed marble. In both cases the jigs were allowed to achieve steady-state and operation was interrupted, the whole laboratory jig bed was sliced and analyzed so that the stratification profiles could be measured.

Optimum values of the specific stratification constant were determined from the best fit between the equilibrium stratification model given by Equations (18) and (19) and measurements of solids yield and recovery of components. These results are summarized in Table I. The model provided very satisfactory comparisons with the experimental data as can be seen in Figures 4 and 5 and Tables II and III. The model appears to fit the data for the heavy fraction less well than the data for the two lighter fractions as seen in Table III and Figure 5. This is probably due to the variation in particle shape and surface roughness between coal and the broken marble which would lead to variations in the value of α for the different materials. In this section and the next, yield of solids and recovery of components to the heavy product (refuse) are recorded. This is only a matter of convenience.

The relatively high value of α for the PVC cubes indicates that this material was separated more efficiently than the coal particles which were significantly more variable

TABLE I

Experimental conditions examined by Vetter[7] in a batch jig.

Material	Size (mm)	Component densities (kg/m^3)			Feed volume composition			Estimated value of the specific stratification constant α
PVC cubes	3.5	1200	1300	1500	0.33	0.33	0.34	0.113
Synthetic mixtures of coal and marble	0.85×0.43	1350	1450	2850	0.5	0.3	0.2	0.098

TABLE II

Measured and calculated component concentrations by layer for batch jigging of PVC cubes. (Duplicate experimental measurements[7] are shown in brackets.)

	Relative density		
Layer	1.2	1.3	1.5
1	0.829 (0.870/0.796)	0.171 (0.108/0.192)	0.000 (0.022/0.012)
2	0.179 (0.199/0.220)	0.734 (0.746/0.704)	0.086 (0.055/0.077)
3	0.001 (0.002/0.000)	0.093 (0.068/0.085)	0.906 (0.930/0.915)

TABLE III

Measured and calculated component concentrations by layer for batch jigging of synthetic mixtures of coal and marble. (Duplicate experimental measurements[7] are shown in brackets.)

	Relative density		
Layer	1.35	1.45	2.85
1	0.991 (0.973/0.978)	0.009 (0.027/0.018)	0.000 (0.000/0.004)
2	0.953 (0.923/0.977)	0.047 (0.067/0.022)	0.000 (0.010/0.002)
3	0.804 (0.700/0.708)	0.196 (0.268/0.279)	0.000 (0.032/0.012)
4	0.463 (0.315/0.329)	0.537 (0.615/0.643)	0.000 (0.069/0.028)
5	0.142 (0.137/0.175)	0.727 (0.528/0.541)	0.131 (0.334/0.284)
6	0.000 (0.011/0.005)	0.002 (0.014/0.012)	0.998 (0.975/0.983)

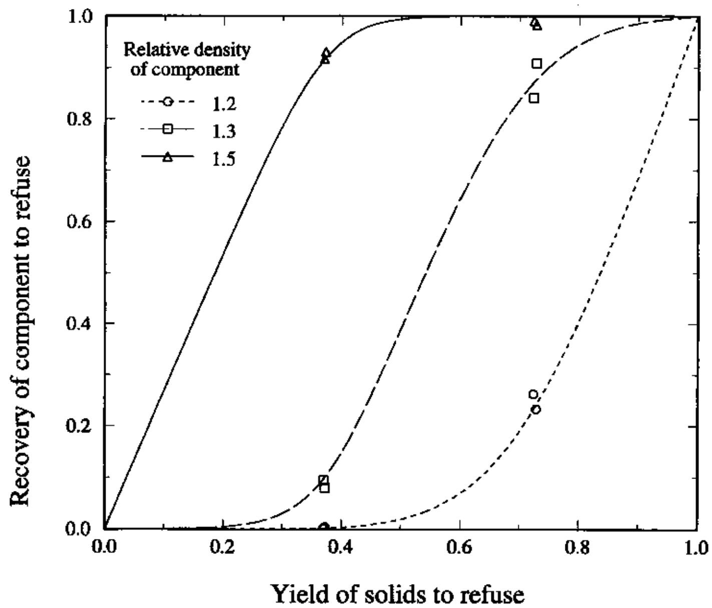

FIGURE 4 Comparison between the component recoveries measured by Vetter[7] and the model predictions for a batch jig operating with synthetic PVC cubes. The points represent the experimental data and the lines the model responses.

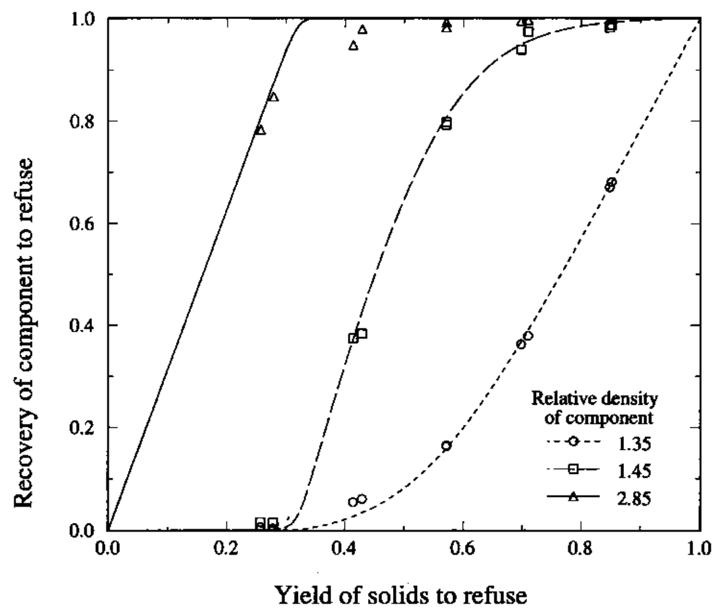

FIGURE 5 Comparison between the recoveries measured by Vetter[7] and the model predictions for a batch jig operating with sized coal. The points represent the experimental data and the lines the model response.

in shape and finer in size than the PVC cubes. This illustrates that the specific stratification constant varies with the shape and size of the particles as well as with the mechanics of the jig mechanism. However α does not vary with the density of the particles.

Comparison of the Model to Continuous Jigging Data

Operating data on two single and five multi-compartment Baum and Batac jigs processing a variety of coals have been compiled from the literature and were used in this work to assess the applicability of the model to industrial jigging operations. The size and washability distributions of the feed to each of the seven jigs are given in Figures 6 and 7.

In this study the effect of size distribution was not considered and the monosize multicomponent version of the dispersion model was applied to the composite data. Considering the lack of information on the transport characteristics of the various jigs, the empirical velocity model given by Equation (25) was used with parameter $\kappa = 0.3$. In addition, as no quantitative information is available on the accuracy of splitting the bed layers, the parameter δ was arbitrarily set to 0, corresponding to a very clean cut at the discharge gate.

Optimum values of the specific stratification constant were determined for each jig from the best fit between the measured and calculated component recoveries from Equation (27), given the experimentally determined yield of solids. The result for the primary separation in Baum jig 1 are shown in Figure 8, where the calculated recoveries of each of the eight washability fractions are shown as a function of the total yield of

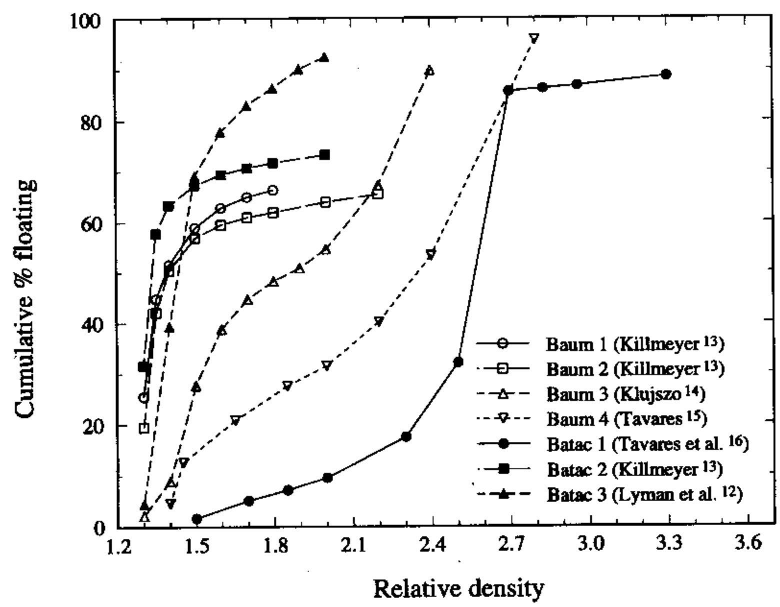

FIGURE 6 Washabilities of the feeds that were processed in the industrial jigs.

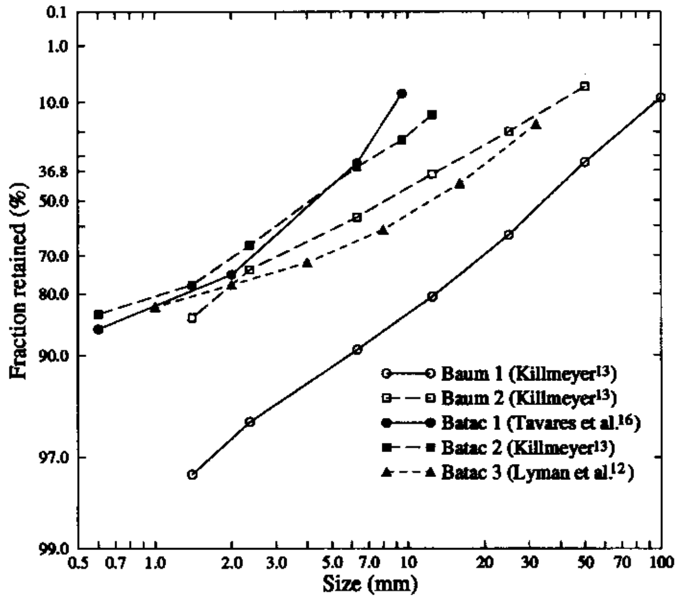

FIGURE 7 Size distributions of the feeds that were processed in the industrial jigs.

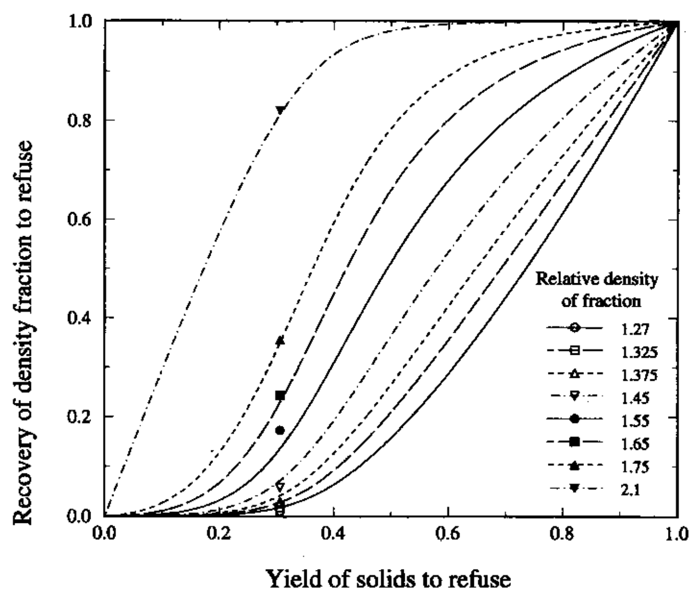

FIGURE 8 Comparison between the recoveries of eight washability fractions measured in the operating jig (points) and calculated by the model (lines) for the separation of coal in the primary compartment of Baum jig 1. Recoveries and yield to refuse are shown.

coal to refuse. In an operating jig, the total yield is controlled primarily by the vertical position of the refuse gate. Raising the gate increases the yield to refuse and *vice versa*. The curves were fitted to the data at the experimentally determined total yield using a single value of α for each compartment. The results for each of the seven data sets are summarized in Table IV.

The specific stratification constants found in these industrial jigs are generally lower than the values found for the Vetter's batch jig and this is to be expected since the batch jig can be operated under ideal conditions. It can also be observed that the stratification constant values increase consistently from the first to the second compartment for each particular jig. This can be explained by more efficient stratification in this second compartment due to reduced bed depths allowing improvements in the bed mobility. However, lower values of α in the primary compartment could also indicate that the residence time in the jigs was not sufficient for the development of the equilibrium stratification profile by the time the moving bed reaches the end of the primary compartment.

Although Figure 8 gives a complete representation of the separating characteristics of the primary compartment of the jig, the data are more conventionally represented as a partition function at a particular setting of the bed splitter. The data from Figure 8 are plotted as a partition curve in Figure 9. Although the partition curve of Figure 9 is the traditional representation of separation performance, Figure 8 contains far more information since it shows how the partition function could be calculated at any total solids yield and therefore discharge gate position. The performance of the secondary compartment could also be represented as in Figure 8 but it is convenient to show just the effective partition curves for both stages as in Figure 9.

The operating data and the corresponding model fit for each of the other industrial jigs are shown in Figures 10 to 15. The model calculations were made using the values of α listed in Table IV. The model is obviously capable of describing the experimental

TABLE IV

Application of the multicomponent continuous jig model.

Jig	Size (mm)	Separation	Measured yield (%)	Estimated value of α	Calculated cut height h_s	Calculated bed density at the cutter position $\bar{\rho}(h_s)$
Baum 1	150×1.4	Primary	30.6	0.027	0.56	1.78
		Secondary	13.1	0.091	0.39	1.56
Baum 2	100×1.4	Primary	44.5	0.046	0.65	1.45
Baum 3	50×1	Primary	20.6	0.009	0.49	2.10
		Secondary	30.1	0.028	0.56	1.95
Baum 4	50×0.6	Primary	5.0	0.006	0.27	2.56
		Secondary	71.7	0.114	0.83	1.77
Batac 1	12×0.6	Primary	7.4	0.019	0.30	3.40
		Secondary	96.4	0.113	0.98	1.68
Batac 2	19×0.6	Primary	14.8	0.028	0.39	2.23
		Secondary	17.3	0.072	0.42	1.66
Batac 3	50×0.6	Primary	15.2	0.068	0.42	1.70

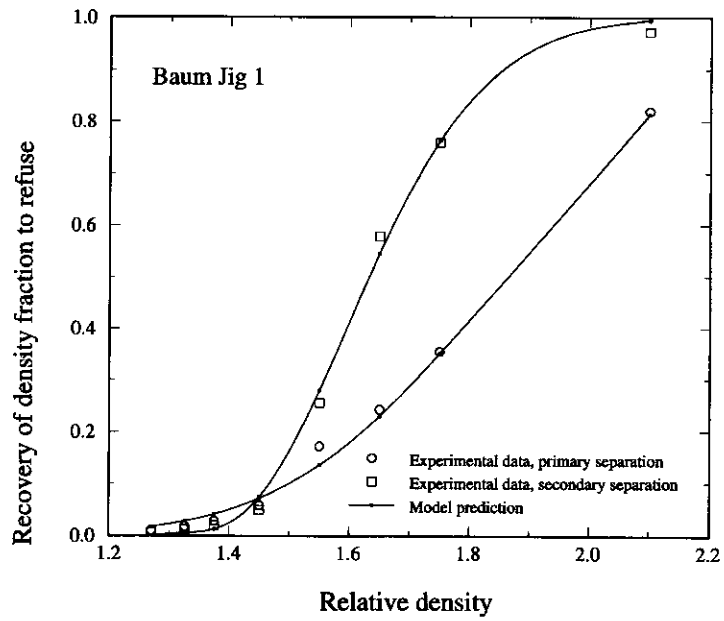

FIGURE 9 The effective partition curves for the primary and secondary compartments of Baum jig 1 calculated using the data of Figure 8 at the measured solids yield in each compartment.

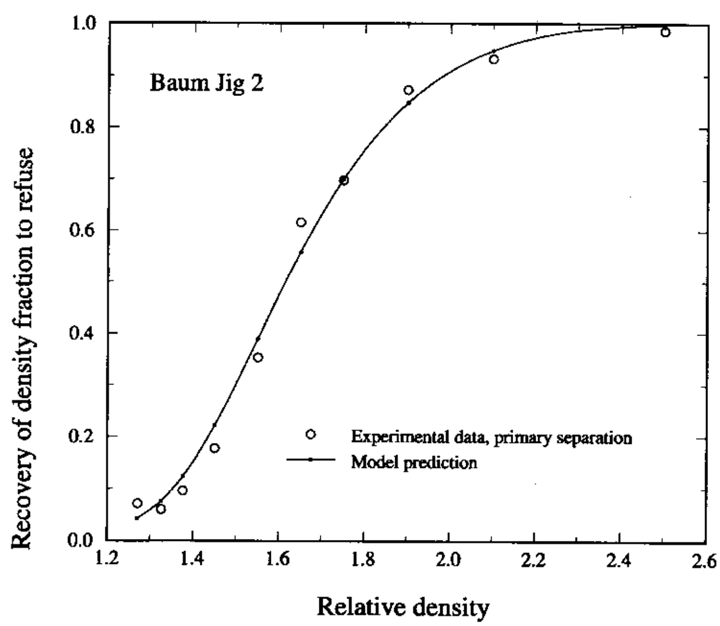

FIGURE 10 The effective partition curve for Baum jig 2 measured and calculated from the model at the experimentally determined solids yield.

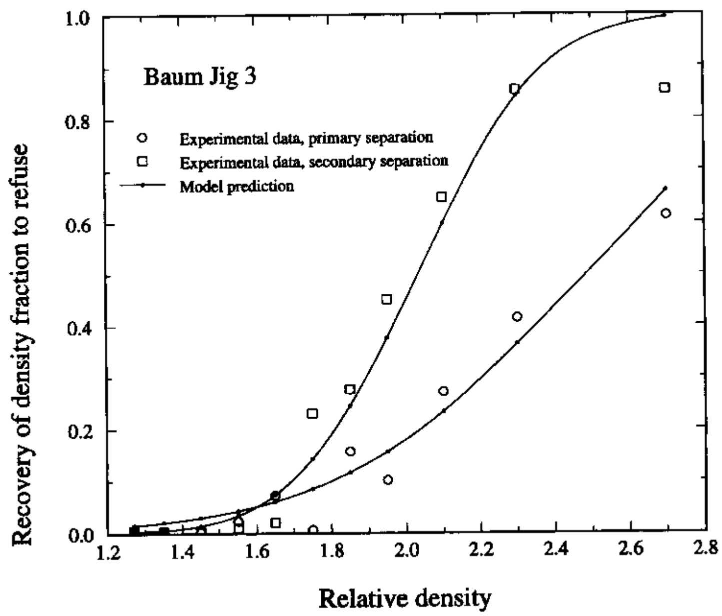

FIGURE 11 The effective partition curves for the primary and secondary compartments of Baum jig 3 showing the comparison between the model and the measured performance.

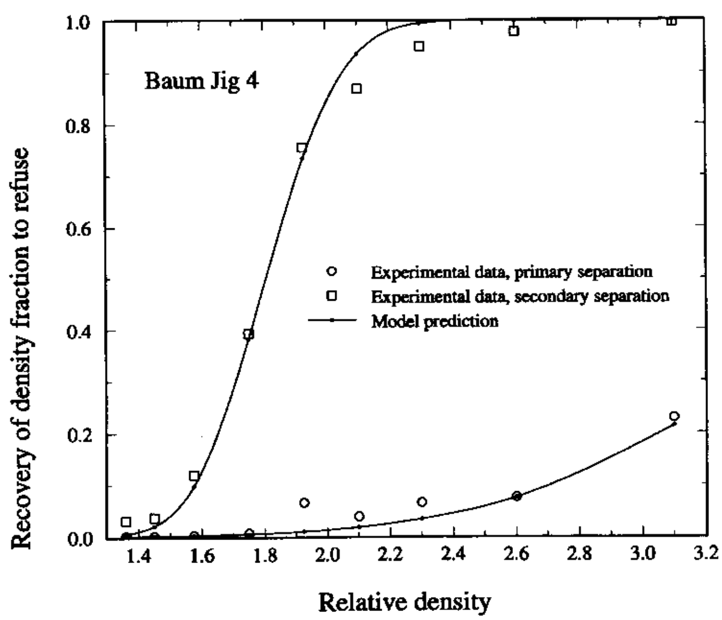

FIGURE 12 The effective partition curves for the primary and secondary compartments of Baum jig 4 showing the comparison between the model and the measured performance.

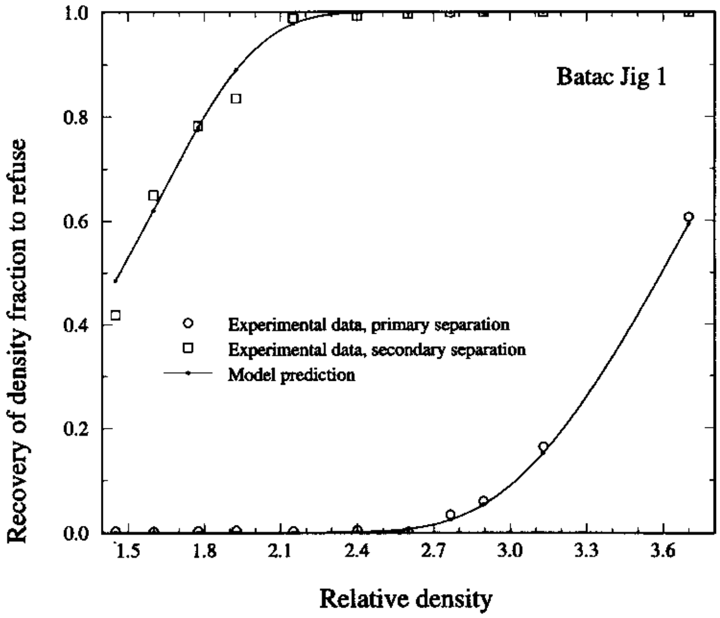

FIGURE 13 The effective partition curves for the primary and secondary compartments of Batac jig 1 showing the comparison between the model and the measured performance.

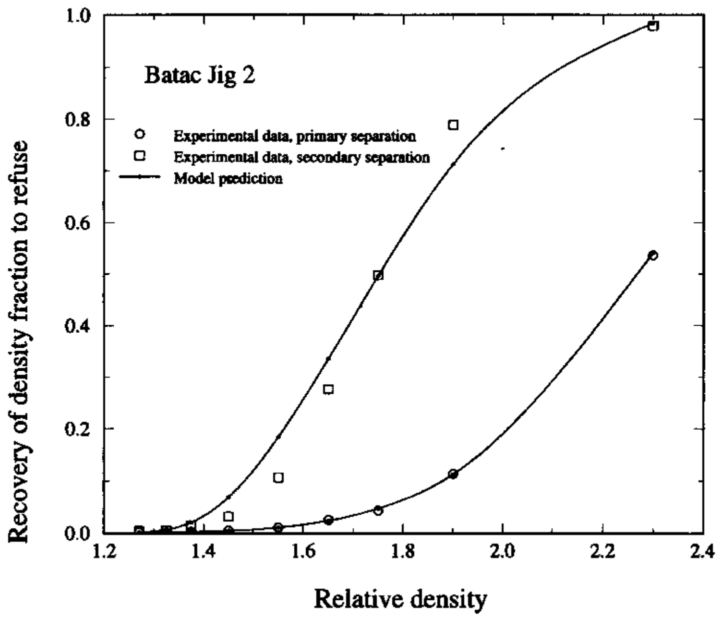

FIGURE 14 The effective partition curves for the primary and secondary compartments of Batac jig 2 showing the comparison between the model and the measured performance.

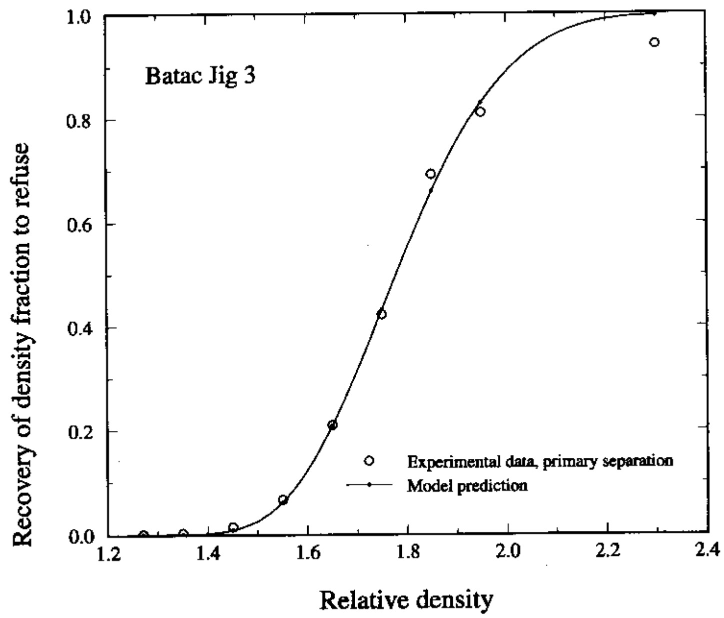

FIGURE 15 The effective partition curve for Batac jig 3 measured and calculated from the model at the experimentally determined solids yield.

data to within the experimental errors present in the data.

The stratification profiles in the operating jig beds are also interesting although the profiles were, of course, not measured during the operation of these industrial jigs. It is difficult, though not impossible, to measure the profiles in industrial equipment, and the development of this model will provide some incentive to do so since that is the best test of the validity of the model. The calculated stratification profiles for each compartment in each jig are shown in Figures 16 to 27. The profiles are seen to differ widely particularly due to the variation in feed washabilities, as observed in Figure 6. Even jigs that have comparable values of α such as Batac 1, Batac 2 and Baum 1 have widely differing equilibrium stratification profiles because of the differing make-up of the feed. Also shown in the figures are the estimated cut height to match the experimentally determined partition data.

The average density profile $\overline{\rho}(h)$ in the jig bed calculated from the dispersion model using Equation (11) is also interesting and important particularly for automatic control of the discharge gate position. The bed density at the split height for each jig and compartment is presented in Table IV. The rate of shale and refuse discharge from an operating jig is usually controlled by a density-sensitive float[2] and the vertical position of the float is governed by the average density profile. The model can be used to calculate the sensitivity of product density to float position and consequently can be used to design effective control systems for the operating jig.

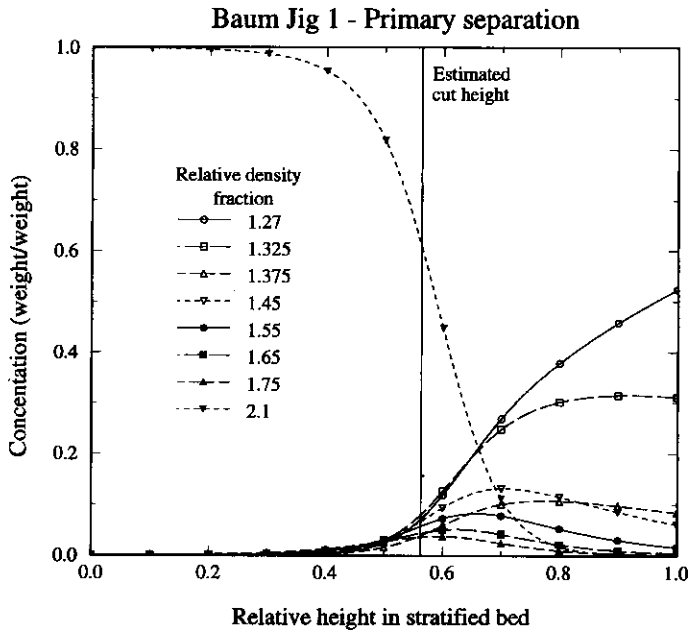

FIGURE 16 Calculated stratification profiles in the operating bed for the primary compartment of Baum jig 1. The calculated position of the cutter is shown as a vertical line.

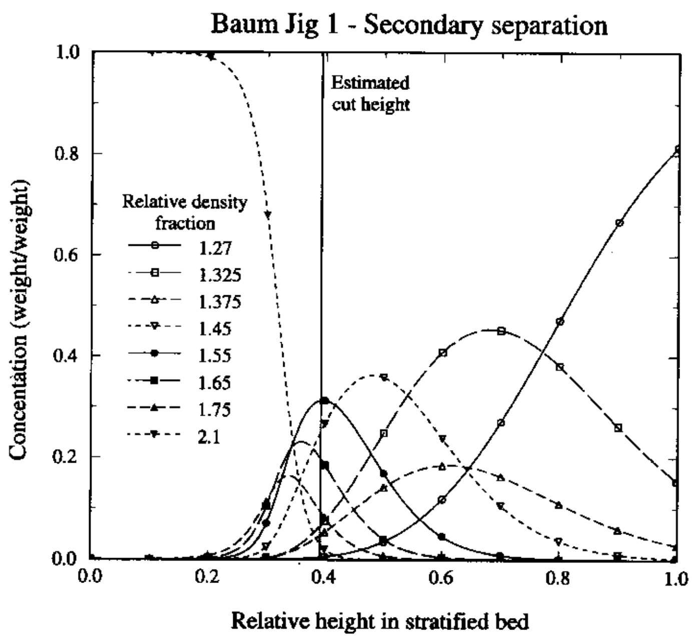

FIGURE 17 Calculated stratification profiles in the operating bed for the secondary compartment of Baum jig 1. The calculated position of the cutter is shown as a vertical line.

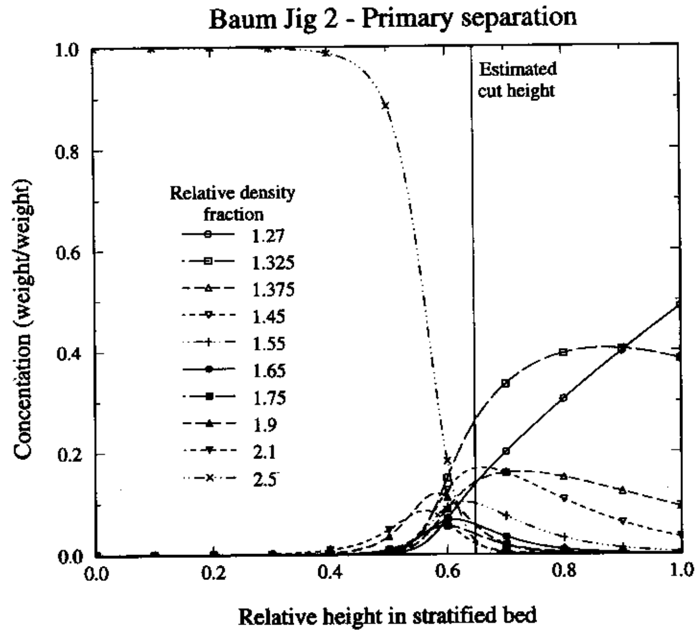

FIGURE 18 Calculated stratification profiles in the operating bed of Baum jig 2. The calculated position of the cutter is shown as a vertical line.

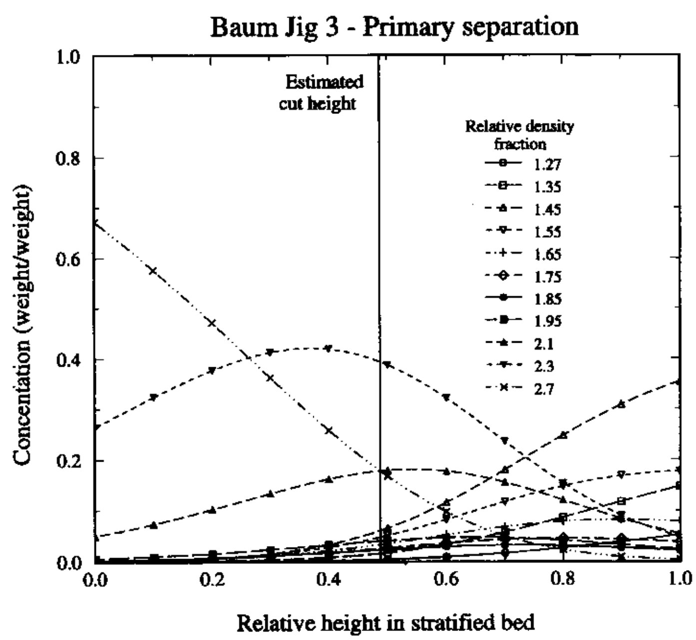

FIGURE 19 Calculated stratification profiles in the operating bed for the primary compartment of Baum jig 3. The calculated position of the cutter is shown as a vertical line.

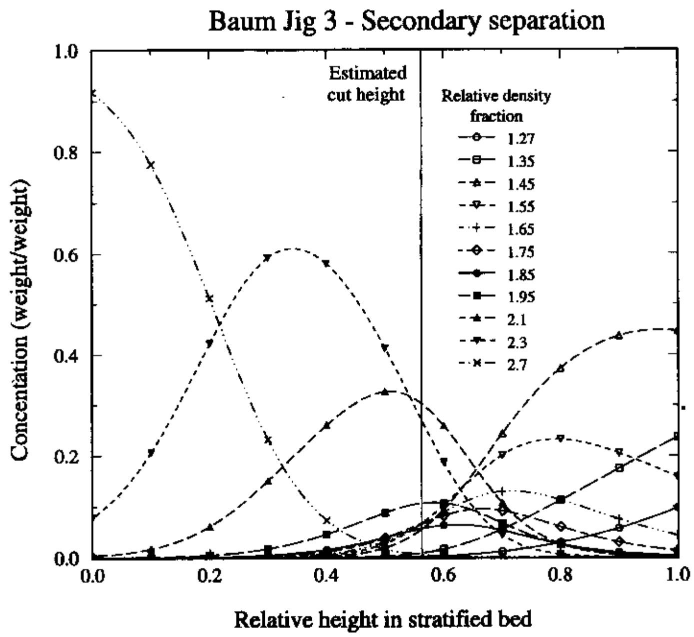

FIGURE 20 Calculated stratification profiles in the operating bed for the secondary compartment of Baum jig 3. The calculated position of the cutter is shown as a vertical line.

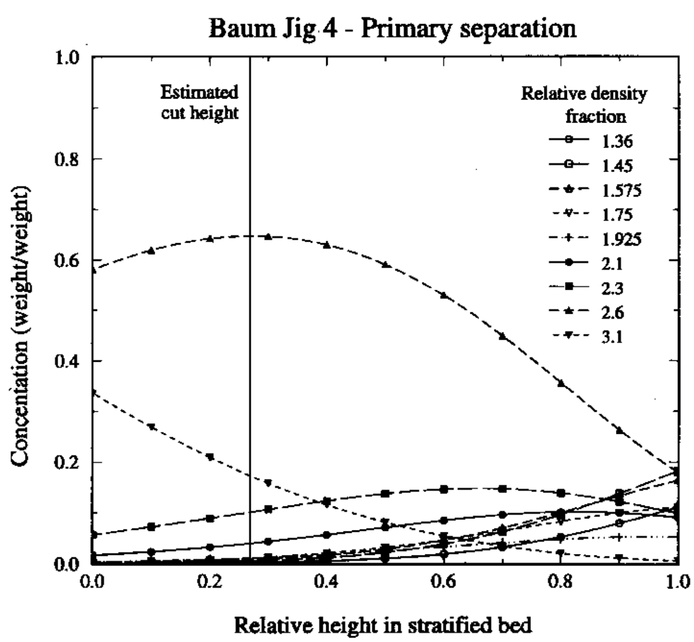

FIGURE 21 Calculated stratification profiles in the operating bed for the primary compartment of Baum jig 4. The calculated position of the cutter is shown as a vertical line.

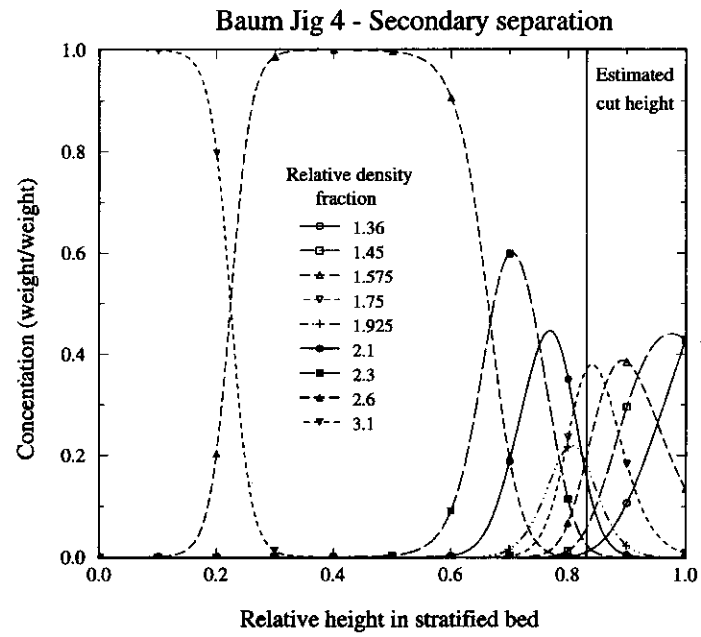

FIGURE 22 Calculated stratification profiles in the operating bed for the secondary compartment of Baum jig 4. The calculated position of the cutter is shown as a vertical line.

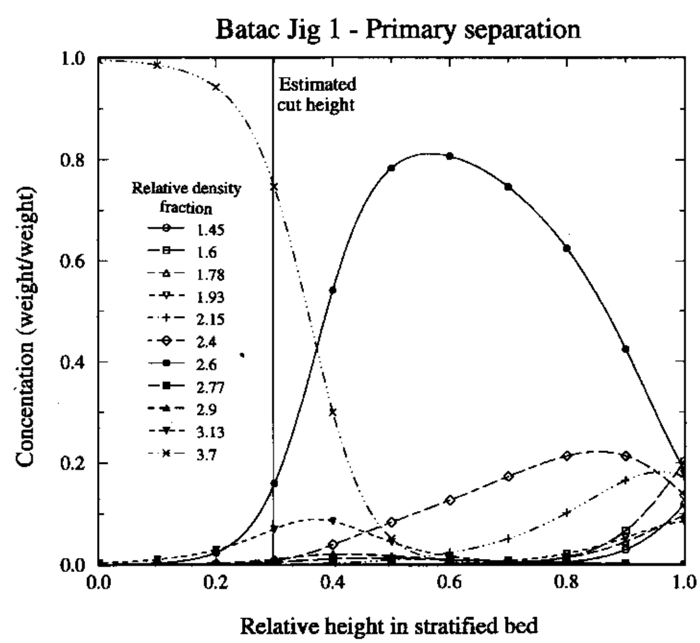

FIGURE 23 Calculated stratification profiles in the operating bed for the primary compartment of Batac jig 1. The calculated position of the cutter is shown as a vertical line.

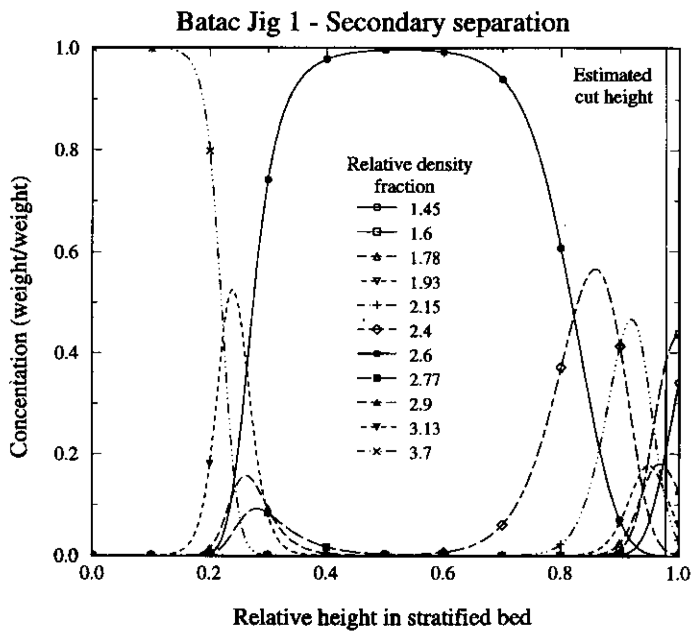

FIGURE 24 Calculated stratification profiles in the operating bed for the secondary compartment of Batac jig 1. The calculated position of the cutter is shown as a vertical line.

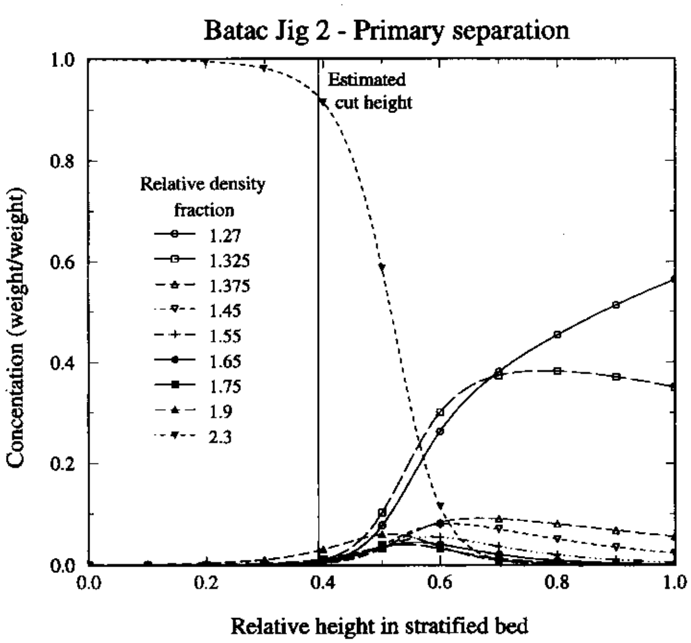

FIGURE 25 Calculated stratification profiles in the operating bed for the primary compartment of Batac jig 2. The calculated position of the cutter is shown as a vertical line.

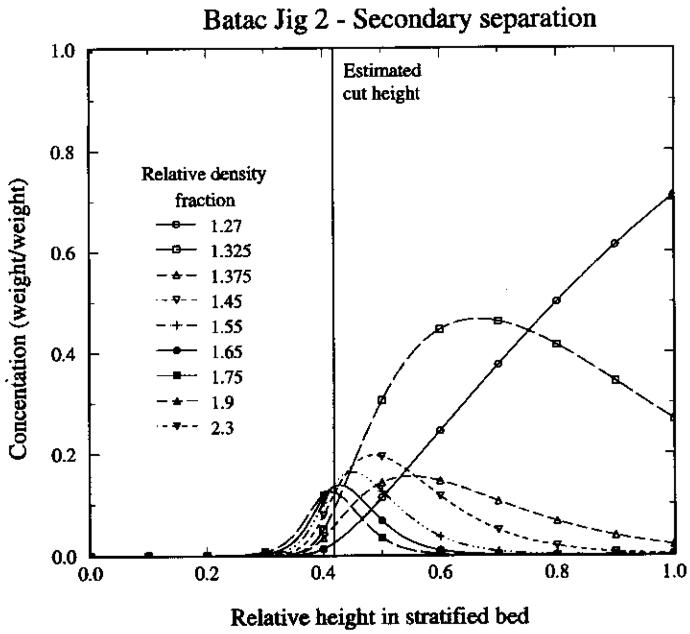

FIGURE 26 Calculated stratification profiles in the operating bed for the secondary compartment of Batac jig 2. The calculated position of the cutter is shown as a vertical line.

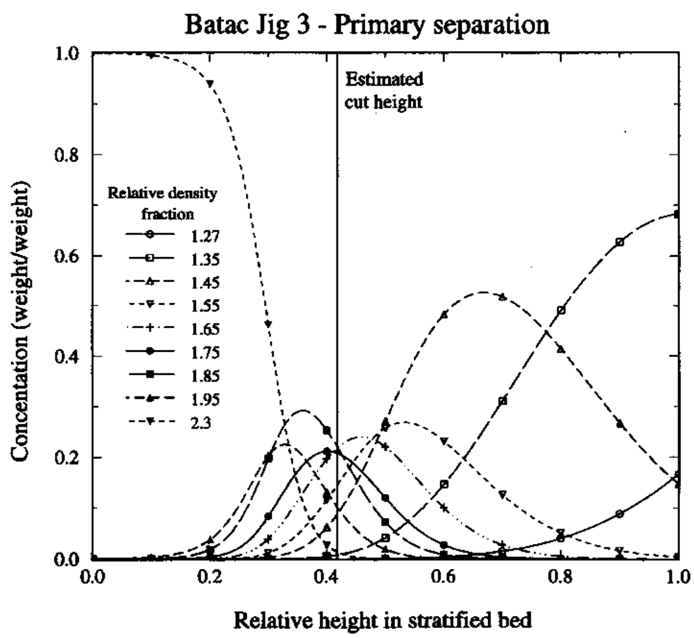

FIGURE 27 Calculated stratification profiles in the operating bed of Batac jig 3. The calculated position of the cutter is shown as a vertical line.

DISCUSSION

Several researchers have recognized the need to directly measure and model bed stratification (Armstrong and Wallace[17], Rong[18], Lyman *et al.*[12]). In a recent paper, Lyman *et al.*[19] developed an interpolation method to describe stratification in jigging beds. Although useful in producing smooth curves from the experimental data, as any interpolation method it produces non-natural fitting to noisy data. From the interpolated concentration profiles these researchers have calculated an index called jigging stratification index (JSI). While suited to evaluate the accuracy in the bed stratification, it can not be used in the prediction of jig performance as the washability of the feed changes since, without a model such as the one developed here, it is not possible to calculate the variation of JSI with feed washability. On the other hand, the specific stratification constant presented in this paper can be used for prediction since it is constant and independent of the washability of the feed.

The traditional procedure of using partition curves for performance evaluation and simulation of jigs is highly artificial. Although the partition curve gives a single measure of the overall jigging process, it does not allow examination of each of the separate processes: stratification, transport and separation of product and refuse layers. The approach used in this paper provides a framework by which the industrial jigging process can be analyzed and effectively modeled. The model also emphasizes the importance of the velocity profile and provides an incentive to measure this profile in industrial jigs.

The main advantage of the model is that only a single parameter, α, must be estimated from experimental data. This contrasts with empirical models based on the partition curve or multilinear regressions which have many parameters. Dimensionless quantities (bed height and longitudinal velocity) are used in the model, no specific information on each jig is required to apply the model. The value of α gives an immediate indication as to how well the jig is operating. Well operated jigs have high values of α while excessively low values of α indicate poor and inefficient operation.

Although derived from Mayer's potential theory idea, the major drawbacks of the former are remediated and the model is based on the concept of equilibrium stratification. The model in its present form can be applied strictly to monosize feed only. When the feed has a wide range of particle sizes, the calculation of the potential energy is far more complicated than shown in Equation (1). The development of the model for size distributed systems will be the subject of a future paper. The model developed in this paper is available in the MODSIM coal and mineral processing plant simulator (King[20]).

It will be important to establish the relationship between the value of the specific stratification constant, α, and the specific operating conditions in the jig such as pulse shape and frequency and other mechanical features of the machine. α can be readily calculated from the experimentally measured recoveries in an operating jig. It should be possible to establish the necessary relationships by careful experimentation.

The equilibrium stratification model was inspired originally by Einstein's[21] analysis of the Brownian Motion and especially by Perrin's[22] subsequent work on the stratification of small particles suspended in water in the gravitational field. Equation (6) is a direct analogy to Perrin's equation[23]

$$-\frac{2}{3}W\,\frac{dC}{dH} = gV_p\,C_p\,(\rho - \bar{\rho})$$

where W is the mean kinetic energy of a particle in the bed. The representation of the stratification profile in Figure 1 was inspired by Perrin's remarkable photograph of the equilibrium distribution of small gamboge (0.6 μm diameter) particles[22]. Dr. J. W. Evans[24] has pointed out that by analogy to the Einstein equation

$$\mu = \frac{RT}{N} \frac{1}{3\pi D d_p}$$

the model provides a definition of the "viscosity" of the stratifying bed and this could possibly form the basis for the calculation of a consistent velocity profile in the continuous jig. These and other interesting analogies with molecular theory will be examined in a future paper.

CONCLUSIONS

A fundamental model based on the potential energy minimization principle allowing for dispersive effects is presented and applied to the batch jigging of coal. A coupled transport/stratification model was used to describe the continuous jigging of coal. The model has been applied to the jigging of various coals in industrial Baum and Batac jigs. The model describes these operations exceptionally well. The model requires only one parameter which has a detinite experimental significance and which can be calculated from operating jig performance data

Through the use of dimensionless quantities (bed height and axial velocity) the model can be applied to a wide variety of systems without the need to know specific operation and design characteristics.

The stratification model can be used directly to predict the variations of product yield and quality with the variation in discharge gate position. The effect of variation of feed composition on the bed .stratification can be correctly predicted and therefore the model can be used for reliable simulation of coal washing plants.

Acknowledgements

The authors acknowledge fruitful discussions with Dr. G. J. Lyman who generously made available to us his extensive data on the operation of pilot-scale batch and continuous jigs. These data, which are very comprehensive, include measured stratification profiles in size distributed beds and will be analyzed in a future publication. It was Dr. Lyman who first pointed out to us the significance of the velocity profile in the continuously operating jig.

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List of Symbols

C_i = volume fraction of particles in washability fraction i

C_ρ = volume fraction of particles of density ρ at height h in the bed

d_p = particle size m

D = dispersion coefficient $m^2\ s^{-1}$
E = potential energy of the bed J
g = gravitational constant $m\ s^{-2}$
h = relative height in stratified bed
h_s = split relative height in the bed
H = height in stratified bed m
H_b = height of the bed m
n_D = diffusive flux $m^3\ s^{-1}\ m^{-2}$
n_S = stratification flux $m^3\ s^{-1}\ m^{-2}$
N = Avogadro's constant mol^{-1}
R = universal constant of gases J $mol^{-1}\ K^{-1}$
R_i = recovery of washability fraction i to clean coal
T = absolute temperature K
u = specific mobility $m\ N^{-1}\ s^{-1}$
v = dimensionless longitudinal velocity at height h in the bed
V = longitudinal velocity at height h in the bed $m\ s^{-1}$
V_p = volume of a particle m^3
Y = total yield of solids to clean coal
α = specific stratification constant $m^3\ kg^{-1}$
μ = coefficient of viscosity $kg\ m^{-1}\ s^{-1}$
ρ = particle density $kg\ m^{-3}$
$\bar{\rho}$ = average density of particle at height h $kg\ m^{-3}$

References

1. A. M. Gaudin, *Principles of Mineral Dressing*, McGraw-Hill, New York, 1939, Chapter 12.
2. H. L. Lovell, R. G. Moorehead, P. T. Luckie and J. K. Kindig, Chapter 7.1.2 in *Coal Preparation*, (J. W. Leonard III, ed.) 5th ed., Society of Mining Engineers, Littleton, 1991, pp. 301–367.
3. F. W. Mayer, "Fundamentals of a Potential Theory of the Jigging Process," *Proceedings 7th International Mineral Processing Congress* (N. Arbiter, ed.) Gordon and Breach, New York, **1**, 1964, pp. 75–86.
4. G. J. Lyman, "Review of Jigging Principles and Control", *Coal Preparation*, **11**, pp. 145–165 (1992).
5. K. F. Tromp, "New Methods of Computing the Washabilities of Coals," *Glückauf*, **73**, pp. 125–131, 151–156 (1937).
6. B. S. Gottfried,"A Generalization of Distribution Data for Characterizing the Performance of Float-Sink Coal Cleaning Devices," *International Journal of Mineral Processing*, **5**, pp. 1–20 (1978).
7. D. A. Vetter, *Mathematical Model of a Fine Coal Batch Jig*, M.Sc. Thesis, University of Natal, Durban, 1987.
8. F. F. Peng and P. T. Luckie, Chapter 10.1 in *Coal Preparation*, (J. W. Leonard III, ed.) 5th ed., Society of Mining Engineers, Littleton, 1991, pp. 659–716.
9. R. P. King, "A Quantitative Model for Gravity Separation Unit Operations that Rely on Stratification," *APCOM 87*, Proceedings of Twentieth International Symposium on the Application of Computers and Mathematics in the Mineral Industries, Johannesburg, SAIMM, **2**, 1987, pp. 141–151.
10. J. A. Davis and R. L. Whitmore, "An Experimental Analysis of the Separation Characteristics of a Commercial Baum Jig," *Proceedings 4th International Coal Preparation Congress*, 1962, Paper E5, pp. 329–339.
11. R. X. Rong, C. J. Wood and D. M. Hughes, "Analysis and Improvement of Jig Performance," *Proceedings 4th Australian Coal Preparation Conference*, Gladstone (1988) pp. 133–149.
12. G. J. Lyman, A. Jonkers, L. van Latum and D. M. Hughes, "Computerized Automatic Control of Coal Washing Jigs," Final Report, NERDDP Project 1028 (1990).
13. R. P. Kilmeyer, Jr., "Performance Characteristics of Coal-Cleaning Equipment: Baum and Batac Jigs", Report of Investigations, PMTC-9(80), USDOE (1980).

14. L. A. C. Klujszo, *Estudo para a Utilização de Espirais de Reichert LD9 no Circuito de Finos do Lavador do Recreio, Copelmi,* M.Sc. Thesis, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil, 1993.
15. L. M. Tavares, *Modelamento e Simulação do Beneficiamento Gravimétrico de Carvão Mineral,* M.Sc. Thesis, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil, 1991.
16. L. M. Tavares and J. Rubio, "Performance Evaluation and Simulation of a Batac Jig Cleaning Pyrite from Coal Washery Tailings," in *Processing and Utilization of High-Sulfur Coals IV,* (P. R. Dugan, D. R. Quigley and Y. A. Attia, eds.) Elsevier Science Publishers, 1990, pp.597–607.
17. F. Armstrong and W. M. Wallace, "Diagrammatic Representation of Jig Washing," *Proceedings 2nd Symposium on Coal Preparation,* University of Leeds, 1957, pp. 417–433.
18. R. X. Rong, *Fundamental Studies of the Stratification Mechanism of the Jig,* Ph.D. Thesis, Julius Kruttschnitt Mineral Research Centre, University of Queensland, 1990.
19. G. J. Lyman, L. van Latum, A. Jonkers and D. M. Hughes, "Application of Line-Length Interpolation Methods to Problems in Coal Preparation – II: Bulk Density, Size Distribution and Stratification Index for a Jig Bed," *Coal Preparation,* **13** (1993) pp. 157–178.
20. R. P. King, "Optimization of Coal Washing Plants by Simulation," to be published (1995).
21. A. Einstein, "On the Movement of Small Particles Suspended in a Stationary Liquid Demanded by the Molecular-Kinetic Theory of Heat," *Annales de Chemie et de Physique,* **17** 549 (1905). English Translation in Investigations on the Theory of the Brownian Movement (R. Fürth and A. D. Cowper, eds.) Methuen, London (1926).
22. M. J. Perrin, "Brownian Movement and Molecular Reality," *Annales de Chemie et de Physique* **8** eme Ser. (1909). English Translation by F. Soddy, Taylor and Francis, London (1910).
23. Ibid., p. 24.
24. J. W. Evans, Personal Communication, 1992.

Industrial Washabilities.txt

Data for Figure 6.

Baum 1	
1.30	25.4
1.35	44.8
1.40	51.0
1.50	58.9
1.60	62.6
1.70	64.9
1.80	66.1
Baum 2	
1.30	19.7
1.35	42.2
1.41	50.4
1.50	56.9
1.60	59.3
1.70	60.9
1.80	61.8
2.00	63.9
2.20	65.3
Baum 3	
1.30	1.85
1.40	8.97
1.50	27.9
1.60	38.9
1.70	44.6
1.80	48.1
1.90	50.6
2.00	54.3
2.20	66.8
2.40	89.5
Baum 4	
1.40	4.78
1.45	12.8
1.65	21.0
1.85	27.8
2.00	31.7
2.20	40.3
2.40	53.3
2.80	95.8
Batac 1	
1.50	1.67
1.70	5.36
1.85	7.51
2.00	9.82
2.30	17.8
2.50	32.2
2.70	85.4
2.83	85.8
2.96	86.4
3.30	88.3
Batac 2	
1.30	31.6
1.35	57.7
1.40	63.5
1.50	66.9
1.60	69.3
1.70	70.6
1.80	71.7
2.00	73.1
Batac 3	
1.30	4.33
1.40	39.0
1.50	68.6
1.60	77.5
1.70	82.6
1.80	86.1
1.90	89.5
2.00	92.0